The Ladies' Diary

The last years, 1835–1840

Twenty problems from the final six issues

Virtue and taste, with female softness join'd,
All that subdues and captivates mankind,
In Britain's matchless fair resplendent shine.

The Diary's title page, 1835

Working draft

A word before the problems

Leybourn's collection stops in 1816. The Diary did not. It ran for another twenty-four years, under Olinthus Gregory of the Royal Military Academy, until the Stationers' Company merged it with *The Gentleman's Diary* in 1841. None of those later questions is in Leybourn.

This booklet is made from one bound volume of the last six issues, for 1835 to 1840, the 132nd to the 137th almanacs "published of this kind". The copy is Harvard College Library's, digitised by HathiTrust, and came to Harvard from the estate of a George Eastwood. A Mr Geo. Eastwood of Dewsbury is among the solvers in 1835; they may be the same man.

Each issue printed the answers to the previous year's fifteen questions and set fifteen new ones, followed by an appendix of "curious and valuable mathematical papers". So a single issue never holds a question together with its answer. The questions in this volume run from 1584 to 1673. The answers to the last fifteen, set in 1840, were printed in 1841 in *The Lady's and Gentleman's Diary*, which we do not have; for those, and for any answer we could not read, the answers here are our own.

By now the solvers were schoolmasters, surveyors, cadets and assistants at the Nautical Almanac Office, with W. S. B. Woolhouse and William Rutherford of Woolwich answering almost everything. They came from further afield, too. Two questions in 1839 were set by "M. Z. Ameen Bey, a scientific Egyptian now studying in England", who appears in 1840 as "Colonel Ameen Bey, Turkish Artillery". Professor Charles Gill of St Paul's College, New York, sent answers and a question.

And there were still women. A "Miss L. L." set one question in 1834 and another in 1836 (problem 18), and her name appears among the solvers again and again in 1837. A Mary Forster answered in 1837 too. In the enigmas and verse, women are everywhere: Winifred Waverton of Lambeth, Mrs and Miss Richardson of Vauxhall, Helen Ogden of Oldham, Frances Plumbe of Bromley, Jane Sherwill of Plympton and many more.

How to read the labels

Q1608 is the Diary's own question number. *After Q1669* means the question has been retold, or its general data replaced by a worked case; the answer at the back is then for the retelling. The bands are those of the first book in this series. These issues have nothing young enough for Green, and their weight is in Purple and beyond.

How this edition was made

This edition was prepared with the assistance of Claude, an AI model made by Anthropic. Under the editor's direction, Claude worked from HathiTrust's machine-read text of the Harvard copy. It drafted the retellings, solutions, notes and introduction, recomputed every answer, and typeset the text. Selection, editorial decisions and final responsibility for the text are the editor's.

Blue

ages 8–11

1. Counting diagonals

Q1629, from Gill's Mathematical Miscellany

How many diagonals can be drawn in a hexagon? In an octagon? Find a rule for a figure with n sides.

Amber

ages 11–14

2. The shilling on the chessboard

Q1597, by W. F.

A shilling is tossed and lands flat on a chessboard. The side of each square is twice the diameter of the shilling. What is the chance that it lies wholly inside one square, touching no line?

3. Three to two

Q1614, by Thomas Todd

In an isosceles triangle, the two equal sides together are as long as the base and the height together. Show that the base is to the height as 3 is to 2.

Pink

ages 14–16

4. The orthocentre and the cotangent *Q1599, by the Rev. John Sampson*

In a triangle ABC the three altitudes meet at P. Show that $PC = AB \cot C$.

5. An area from the incircle *Q1600, by Thomas Baker*

The incircle of a triangle touches side AB and divides it into two pieces. Show that the product of the two pieces and $\cot \frac{1}{2}C$ equals the area of the triangle.

6. Midpoints and contact points *Q1635, by W. F. Harrison*

On each side of a triangle, the distance between the midpoint of the side and the point where the incircle touches it is half the difference of the other two sides. Prove it.

7. The area from three altitudes *Q1605, by W. Rutherford*

The three altitudes of a triangle are p_1, p_2, p_3. Find its area. Work it out for altitudes 12, 15 and 20.

8. The flag and the tower *after Q1669, by W. Rutherford*

A flagstaff a feet high stands on top of a tower b feet high. Where on the level ground must an observer stand, with his eye h feet above the ground, so that the tower and the flagstaff look equally tall? Work it out for $a = 30$, $b = 20$, $h = 5$.

9. Two semicircles in a semicircle *Q1668, by W. S. B. Woolhouse*

Draw a semicircle. With the two ends of its diameter as centres, draw two more semicircles, on the same side, that touch each other. Show that a common tangent to these two also touches the first semicircle.

10. Three equilateral triangles *Q1633, by W. Rutherford*

On each side of a triangle, draw an equilateral triangle outwards. From each corner of the original triangle, drop a perpendicular to the line joining the two outer vertices nearest to it. Show that the three perpendiculars meet in a point.

Purple

ages 16–17

11. Ameen Bey's integral

Q1653, by M. Z. Ameen Bey

Find $\displaystyle\int_0^\infty e^{-x^2}\,dx$.

12. Ameen Bey's series

Q1654, by M. Z. Ameen Bey

Find the nth term, and the sum of n terms, of $1 + 3 + 8 + 24 + 65 + 153 + 320 + 610 + \cdots$.

13. Cylinders without end

Q1663, by John Butterworth

Cut the largest possible cylinder out of a cone. Then cut the largest cylinder out of the small cone left on top, and so on for ever. What fraction of the cone do all the cylinders fill?

14. The shortest quadrilateral in a rectangle

Q1621, by Robert Sumner

Show that among all quadrilaterals with one corner on each side of a rectangle, the least perimeter equals the sum of the two diagonals. Show that there are infinitely many such quadrilaterals, all parallelograms with sides parallel to the diagonals.

15. Circles in a parabola

Q1622, by W. F. Harrison

A string of circles is placed inside a parabola, along its axis, each touching the parabola and the one before. The first touches it at the vertex and has its centre at the focus. Find the radius of the nth circle.

16. The third circle

Q1661, by Philip Beecroft

Two circles touch each other and touch a straight line at the fixed points A and B. A third, small circle touches the line and both circles. As the first two circles change size, keeping their points of contact A and B, find the path of the centre of the third.

17. Equal pressures

Q1641, by Robert Sumner

A rectangle of length a is lowered into water, tilted at any angle, with one short edge just at the surface. Divide it, by lines parallel to that edge, into n strips that bear equal pressures.

18. To Constantinople by the equator

Q1608, by Miss L. L.

A traveller undertakes to go from London to Constantinople and to touch the equator on the way. Treating the Earth as a sphere, find where he should touch it and how long his journey is, if it is to be as short as possible. Take London at latitude $51°33'$, Constantinople at $41°1'$, and the difference of longitude $28°58'$, as the Diary's solver did.

19. The integral test *Q1671, by Amicus*

Let ϕ be positive and decreasing on $x \geq 1$. Show that $\int_1^\infty \phi(x)\,dx$ is finite or infinite according as $\phi(1) + \phi(2) + \phi(3) + \cdots$ converges or diverges.

20. The hollowed cone *Q1589, by James Hann*

A solid cone swings as a pendulum about a horizontal axis through its vertex. A cone with the same base, but with its vertex at the centre of gravity of the first, is hollowed out of it. How much does this change the time of a swing?

Answers

1. 9 in a hexagon and 20 in an octagon. From each corner you can draw a diagonal to every corner except itself and its two neighbours, so $n-3$ of them. That counts each diagonal twice, so there are $n(n-3)/2$.

2. One in four. The shilling lies inside a square exactly when its centre lands at least a radius from every side, that is, inside a smaller square whose side is the square's side less one diameter. Here that is half the side, so a quarter of the area. Mr Mason of Scoulton gave the general rule: with squares n diameters across, the chance is $(n-1)^2/n^2$.

3. Let the equal sides be s, the base b and the height h. Then $2s = b + h$ and $s^2 = h^2 + \frac{1}{4}b^2$. Squaring the first and comparing, $b^2 + 2bh + h^2 = 4h^2 + b^2$, so $2b = 3h$.

4. Let CF be the altitude from C and AD the one from A. Triangles CPD and ABD are similar, because both are right-angled and $\angle PCD = \angle BAD$. So $PC/AB = CD/AD = \cot C$. This is Rutherford's proof; Thomas Hindle and W. F. Harrison gave others. Miss L. L. and Mary Forster are among the solvers listed.

5. With the usual notation the pieces of AB are $s-a$ and $s-b$, and $\cot\frac{1}{2}C = (s-c)/r$. So the product is $(s-a)(s-b)(s-c)/r$. By Heron's formula $(s-a)(s-b)(s-c) = \Delta^2/s$, and $\Delta = rs$, so the product is Δ. W. Hopps proved it with the fact that if three arcs make up a semicircle, the sum of their tangents equals their product.

6. On side BC the contact point is $s-b$ from B, and the midpoint is $\frac{1}{2}a$ from B. Their distance is $|s - b - \frac{1}{2}a| = \frac{1}{2}|c-b|$.

7. Write $u_i = 1/p_i$ and $\sigma = \frac{1}{2}(u_1+u_2+u_3)$. The sides are $2\Delta u_i$, and Heron's formula then gives $1/\Delta = 4\sqrt{\sigma(\sigma-u_1)(\sigma-u_2)(\sigma-u_3)}$. For altitudes 12, 15 and 20 the area is 150: the triangle has sides 25, 20 and 15. Several solvers pointed out that the question was already in Emerson's *Algebra*.

8. The line of sight to the top of the tower must bisect the angle between the lines of sight to its foot and to the top of the flag. By the angle-bisector theorem, the distances from the eye to the foot of the tower and to the top of the flag are as b to a. That gives

$$x^2 = \frac{a^2h^2 - b^2(a+b-h)^2}{b^2-a^2}$$

for the distance x from the tower. With $a = 30$, $b = 20$, $h = 5$ it is about 39.7 feet, where each subtends $27°53'$. With the eye on the ground ($h = 0$) it is $b\sqrt{(a+b)/(a-b)}$, possible only if the flagstaff is taller than the tower. The answer was printed in 1841, in a volume we do not have; this one is ours.

9. Let the first semicircle have centre O and radius R, and the others centres A, B (the ends of the diameter) and radii p, q with $p + q = 2R$. A common tangent is at distances p and q from A and B. Its distance from the midpoint O of AB is the average, $\frac{1}{2}(p+q) = R$. So it touches the first semicircle too. (The answer appeared in 1841; this is ours.)

10. Let A', B', C' be the outer vertices on BC, CA, AB. The circle with centre B' through A also passes through C, and the circle with centre C' through A also passes through B. These two circles meet at A, so their radical axis is the line through A perpendicular to $B'C'$: the first of the three perpendiculars. In the same way the other two perpendiculars are the radical axes of the other pairs among the three circles centred at A', B', C'. The three radical axes of three circles always meet in a point, the radical centre.

11. $\frac{1}{2}\sqrt{\pi}$. Square the integral and write it as a double integral over the quarter-plane, $\iint e^{-(x^2+y^2)}\,dx\,dy$. In polar co-ordinates this is $\int_0^{\pi/2}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta = \frac{\pi}{4}$.

12. The fifth differences are constant, so the nth term is a polynomial of degree five:

$$u_n = \tfrac{1}{60}\big(n^5 + 15n^3 - 90n^2 + 254n - 120\big).$$

The next term is 1,081. The sum of n terms is $\tfrac{1}{360}\,n\big(n^5 + 3n^4 + 25n^3 - 135n^2 + 514n - 48\big)$. Professor Gill of New York found the same differences, 2, 3, 8, 6, 2, and summed by the calculus of finite differences.

13. $\frac{6}{13}$ of the cone. The largest cylinder in a cone fills $\frac{4}{9}$ of it and is cut at a third of the height. The cone left on top has a third of the height and a third of the radius, so $\frac{1}{27}$ of the volume. The cylinders fill $\frac{4}{9}\big(1 + \frac{1}{27} + \frac{1}{27^2} + \cdots\big) = \frac{4}{9} \cdot \frac{27}{26} = \frac{6}{13}$. The first step is Barbara Sidway's question of 1714. (The printed answer appeared in 1841; this one is ours.)

14. Reflect the rectangle in its sides, as a billiard player would. A quadrilateral with one corner on each side unfolds into a path from a point to its image after four reflections, which is a translation by twice a diagonal. So its perimeter is at least that distance, twice the length of a diagonal, which is the sum of the two diagonals. The bound is reached when the path is straight: the sides are then parallel to the diagonals, and any starting point on a side works.

15. Let the parabola be $y^2 = 4kx$, with focus $(k, 0)$. The first circle has radius k. A circle centred on the axis at $(c, 0)$ with $c > 2k$ touches the parabola when its radius satisfies $r^2 = 4k(c - k)$. Writing $r = 2ks$, the condition that each circle touches the next gives $(s_{n+1} - 1)^2 = (s_n + 1)^2$ from the third circle on, and $s_2 = 1 + \sqrt{2}$. So for $n \geq 2$ the radius of the nth circle is $2k\big(2n - 3 + \sqrt{2}\big)$: from the second circle on, the radii go up in equal steps of $4k$, twice the semi-parameter.

16. An arc of a parabola through A and B, with its vertex above the middle of AB, at an eighth of AB. Put $AB = 2d$. Circles of radii r_1, r_2 touching a line and each other have $AB^2 = 4r_1r_2$, so $r_1r_2 = d^2$; the small circle has $1/\sqrt{r_3} = 1/\sqrt{r_1} + 1/\sqrt{r_2}$. Its centre is at height $y = r_3$ and distance $x = 2\sqrt{r_1r_3}$ from A, and eliminating the radii gives $y = x(2d - x)/4d$. (The answer appeared in 1841; this one is ours.)

17. The pressure on a strip is proportional to its area times the depth of its centre, so the pressure on the part from the surface edge down to a distance x along the rectangle goes as x^2, whatever the tilt. The strips bear equal pressures when the dividing lines are at distances $a\sqrt{1/n}, a\sqrt{2/n}, \ldots, a\sqrt{(n-1)/n}$ from the top edge.

18. The shortest route makes equal angles with the equator on the way down and the way up, like light reflected in a mirror. So reflect Constantinople in the equator, to latitude $41°1'$ south, and go by the great circle from London to that image. With the Diary's figures the journey is $95°56'$ of a great circle, about 6,660 miles, and the traveller touches the equator $17°11'$ of longitude east of London. William Rutherford used the equal-angle principle. W. Clough's printed answer is $96°0'45''$, touching at $16°55'$.

19. Because ϕ is decreasing, on each interval $[k, k+1]$ we have $\phi(k+1) \leq \phi(x) \leq \phi(k)$. Integrating and adding, $\phi(2) + \cdots + \phi(n) \leq \int_1^n \phi \leq \phi(1) + \cdots + \phi(n-1)$. So the integral and the series are bounded or unbounded together. This is the Maclaurin–Cauchy integral test. (Amicus's own answer appeared in 1841.)

20. Let the cone have height h and base radius R. Hung by its vertex, the solid cone swings like a simple pendulum of length $(4h^2 + R^2)/5h$. The hollowed cone swings like one of length $3(27h^2 + 8R^2)/110h$. So the time of a swing is multiplied by

$$\sqrt{\frac{3(27h^2 + 8R^2)}{22(4h^2 + R^2)}}.$$

For a cone as wide as it is tall ($R = h$) this is 0.977, so it swings about 2.3 per cent faster. The two times are equal when $R^2 = \frac{7}{2}h^2$, and a wider cone swings more slowly once hollowed. W. S. B. Woolhouse answered it in 1836, by moments of inertia.

When the Diary was wrong

Problem 18. On his own figures, W. Clough's track of $96°0'45''$ is about $4'$ of arc, some five miles, longer than the shortest, and his point of contact is about $17'$ too far west. The editor printed several answers that "differ only in numerical result on account of different assumptions of longitude and latitude". That is true of the others, but not of this difference, which is an error in the working.

Where the Diary simply worked differently

Problem 7. Not an error, but not new: solvers pointed out that the question had already been answered in Emerson's *Algebra*.

Not in this booklet. Question 1623 asked for a proof that a certain sum of rectangles about the centre of gravity of a triangle is constant. The editor printed a note that "some oversight appears to have been committed in the enunciation of this question", and gave instead the true value in terms of the sides.

Source: *The Ladies' Diary* for 1835–1840, Harvard College Library copy, digitised by HathiTrust (`https://hdl.handle.net/2027/hvd.hxvmnt`).